SOLO BUILDER GUIDE

Build the Raph & Remy dashboard and connect every live data source

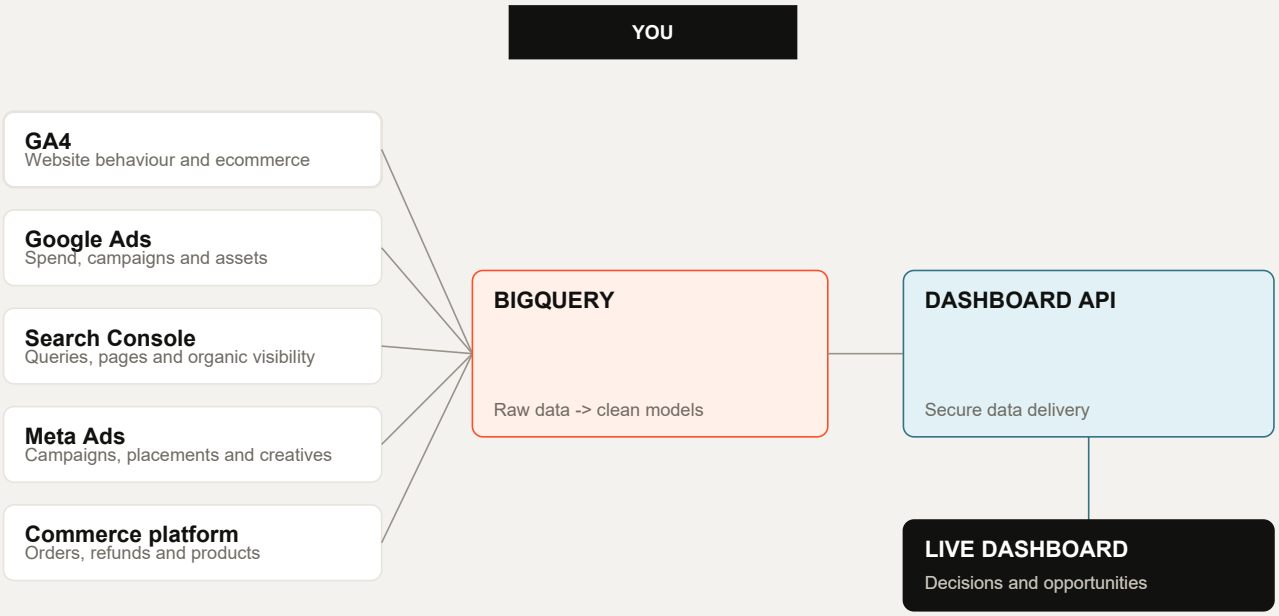
A screen-by-screen implementation manual for one nontechnical person working with Codex. Includes Google Analytics, Google Ads, Search Console, Meta Ads, Shopify, BigQuery and secure deployment.

Solo implementation edition
Version 2.0 | 10 August 2026

READ THIS FIRST

What this guide will help you build

The goal is one private dashboard that refreshes automatically and shows paid media, creative, search, product and revenue performance in one place.



SOURCE	WHAT IT ADDS	NORMAL REFRESH
GA4	Sessions, product views, add-to-cart, checkout and purchases	Daily; optional streaming
Google Ads	Spend, campaigns, ads, conversions and Performance Max	Daily
Search Console	Queries, landing pages, clicks, impressions and ranking	Daily
Meta Ads	Spend, campaigns, placements, actions and creative metrics	Daily
Shopify	Orders, discounts, refunds, products and variants	Daily

HOW TO USE THIS GUIDE

There are only four types of action

Never move to the next page until the completion checks on the current page are true.

YOU

LABEL	WHO ACTS	WHAT IT MEANS
YOU	You	Open the named platform, click the stated controls and record the result.
CODEX	Codex	Paste the supplied prompt into this task and let Codex inspect or change the project.
WAIT	The platform	The connection needs time to create tables. Do not rebuild it while waiting.
CHECK	You and Codex	Compare the dashboard with the source before continuing.

ONE TASK AT A TIME

Complete one numbered action, check the result, then continue. If the screen differs from this guide, take a screenshot and give it to Codex. Do not guess which option is equivalent.

NEVER PASTE A SECRET INTO CHAT

App secrets, access tokens, private keys and passwords belong in a password manager or Google Secret Manager. IDs such as a Google Cloud Project ID, GA4 Property ID and Google Ads Customer ID are safe to record in the worksheet.

CURRENT PROJECT CHECK

The dashboard prototype is not in the current project folder

The workspace inspected for this guide contains a generic web starter. The prototype HTML must be added before Codex can connect its visual components to live data.

YOU

WHAT IS MISSING

No file named raph-remy-growth-dashboard-prototype.html, and no equivalent dashboard implementation, was found in the available project folder.

1. **Find the latest prototype HTML**
   Use the file you previewed at 127.0.0.1:4173. Confirm it contains the current Inter typography and no client-view or analyst-view switch.
2. **Attach it to this Codex task**
   Drag the HTML file into the task, or place it in the current project folder.
3. **Tell Codex that this is the approved visual prototype**
   Do not ask for data integration yet. First ask Codex to inspect and preserve the approved layout.
4. **Keep an untouched copy**
   Name the copy raph-remy-dashboard-prototype-backup.html.

PASTE INTO CODEX AFTER ADDING THE FILE

Find the Raph & Remy dashboard prototype in this workspace. Do not change it. Confirm the filename, list every dashboard page and identify where its dummy data is stored. If the prototype is missing, stop and tell me exactly what file you need.

DO NOT CONTINUE UNTIL

CHECK 1: Codex confirms the exact prototype filename.
CHECK 2: An untouched backup exists.
CHECK 3: Codex lists the dashboard pages without changing the design.

IMPLEMENTATION MAP

Build in small, testable phases

There is no fixed six-day promise. Account access and Meta authorization can add waiting time. Complete the phases in order.

YOU

PHASE	OUTCOME	CONTINUE WHEN
1. Prepare	Project, billing controls, IDs and datasets	All access checks pass
2. Google data	GA4, Google Ads and Search Console in BigQuery	Each source reconciles
3. Paid social	Meta Ads in BigQuery	Spend and purchases reconcile
4. Commerce	Shopify orders, refunds and products in BigQuery	Net revenue reconciles
5. Model	Clean reporting views and opportunity rules	SQL tests pass
6. Connect	Prototype reads live API data	Every page handles loading and errors
7. Launch	Private Cloud Run dashboard	Access and refresh checks pass

EXPECTED PACE

Plan for several focused sessions. GA4 may take up to 24 hours to create data, Search Console may take up to 48 hours, and Meta access issues may take longer. Waiting is normal.

MVP FIRST

Do not add AI analysis, historical Search Console loading or advanced attribution until the five live sources refresh reliably.

WORKSHEET

Record the identifiers you will reuse

Write IDs here or in a notes document. Do not record passwords, app secrets or access tokens.

YOU

Google Cloud Project ID	_____ Example: raph-remy-performance-prod
BigQuery location	_____ One location for every dataset
Billing account	_____ Name only, not payment details
GA4 Property ID	_____ Numbers only
GA4 Measurement ID	_____ Begins with G-
Google Ads Customer ID	_____ Ten digits, usually shown with hyphens
Search Console property	_____ Domain or URL-prefix property
Meta Business ID	_____ Numbers only
Meta Ad Account ID	_____ Usually shown as act_ plus numbers
Meta App ID	_____ Safe to record
Shopify shop name	_____ The part before .myshopify.com
Cloud Run region	_____ Prefer the same region as the data

SAVE SECRETS ELSEWHERE

Meta App Secret, Meta token, Shopify Client Secret and any recovery codes must go into a password manager. The PDF worksheet is not a secret store.

ACCESS CHECK

Confirm you can open every required account

Use the same company Google account where practical. Sign in normally and enable two-step verification before creating transfers.

YOU

PLATFORM	REQUIRED ACCESS	WHERE TO CHECK
Google Cloud	Project Owner or equivalent setup rights	console.cloud.google.com > IAM & Admin > IAM
GA4	Editor or Administrator on the property	analytics.google.com > Admin > Access management
Google Ads	Read access; 2-Step Verification enabled	ads.google.com > Admin > Access and security
Search Console	Verified Owner	search.google.com/search-console > Settings > Users and permissions
Meta	Business, app and ad-account access	business.facebook.com/settings
Shopify	Store owner or staff with app-development access	Shopify Admin > Settings > Users

IF AN ACCOUNT OPENS BUT A BUTTON IS MISSING

That normally means the signed-in user lacks permission. Do not create a second property, ad account or store. Ask the existing owner to raise your access.

DO NOT CONTINUE UNTIL

CHECK 1: You can open all six platforms.
CHECK 2: Your Google Ads account uses 2-Step Verification.
CHECK 3: Search Console shows you as an Owner.

PHASE 1

Choose the BigQuery location before creating datasets

A BigQuery dataset location is difficult to change later. Every source and reporting dataset must use the same location.

YOU

QUESTION	ACTION
Does the company require data to remain in Singapore or a specific region?	Ask the CEO or policy owner before creating any dataset.
No data-residency requirement exists	Use one supported location consistently. US multi-region is usually the simplest compatibility choice.
A regional requirement exists	Confirm that GA4, Google Ads, Search Console, Meta and Shopify transfers all support that location.

STOP BEFORE YOU CHOOSE

This is a governance decision, not a design preference. Record the approved location in the worksheet and use it everywhere.

Official guide: BigQuery dataset locations
https://docs.cloud.google.com/bigquery/docs/locations

DO NOT CONTINUE UNTIL

CHECK 1: The location decision is written down.
CHECK 2: The decision owner is known.

PHASE 1

Create or select the Google Cloud project

Use the existing approved project if one is already linked to billing. Do not create duplicate projects.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/
Sign in with the company Google account.

1. **Open the project selector**
   Click the project name in the top bar.
2. **Look for the approved project**
   If it already exists, select it. If not, click New Project.
3. **Name the new project**
   Use Raph Remy Performance Dashboard. Google will create a separate Project ID.
4. **Copy the Project ID**
   Open Navigation menu > IAM & Admin > Settings. Copy Project ID, not Project number.
5. **Confirm billing**
   Open Navigation menu > Billing. The page must show an active linked billing account.

SAVE THIS

Record the Project ID in the worksheet. Every later BigQuery name and deployment command uses it.

DO NOT CONTINUE UNTIL

CHECK 1: The correct project name appears in the top bar.
CHECK 2: The Project ID is recorded.
CHECK 3: Billing is active.

COST CONTROL

Create a billing budget before loading data

A budget sends alerts. It does not automatically stop services, so keep the amount conservative while building.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/billing/budgets
Confirm the correct billing account or project when prompted.

1. **Click Create budget**
   Name it Raph Remy Dashboard Build.
2. **Set the scope**
   Choose the dashboard project only.
3. **Set a monthly amount**
   Use a low amount you are comfortable monitoring during the build.
4. **Set thresholds**
   Add actual-spend alerts at 20 percent, 50 percent, 90 percent and 100 percent.
5. **Confirm email recipients**
   Make sure the address you actively monitor will receive alerts.
6. **Finish**
   Click Save or Finish, depending on the screen shown.

IMPORTANT

Google states that standard budget alerts monitor spending but do not automatically cap it. Review Billing > Reports during the first month.

Official guide: Create budgets and alerts
https://docs.cloud.google.com/billing/docs/how-to/budgets

GOOGLE CLOUD SETUP

Enable the services the dashboard needs

Enabling an API makes the service available. It does not expose your data publicly.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/apis/library
Check that the dashboard project is selected in the top bar.

SEARCH FOR	OFFICIAL SERVICE NAME
BigQuery API	BigQuery API
BigQuery Storage API	BigQuery Storage API
BigQuery Data Transfer	BigQuery Data Transfer API
Cloud Run	Cloud Run Admin API
Cloud Build	Cloud Build API
Artifact Registry	Artifact Registry API
Secret Manager	Secret Manager API
Identity-Aware Proxy	Identity-Aware Proxy API

For each row: type the service name into the search box, open the matching result, then click Enable. If the button says Manage, it is already enabled.

DO NOT CONTINUE UNTIL

CHECK 1: Every listed service shows Manage instead of Enable.

GOOGLE CLOUD SETUP

Create the datasets used by the managed transfers

A dataset is a folder of tables. Use lowercase names with underscores and the approved location.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/bigquery
The page is called BigQuery Studio.

1. **Expand the project**
   In the Explorer panel, click the arrow beside the dashboard Project ID.
2. **Open the project menu**
   Click the three dots beside the Project ID, then Create dataset.
3. **Enter the Dataset ID**
   Create one dataset at a time using the names in the table below.
4. **Choose the location**
   Select the location recorded in the worksheet. Do not mix regions.
5. **Leave advanced options unchanged**
   Do not add table expiration or encryption settings for the MVP.
6. **Click Create dataset**
   Repeat until all five datasets exist.

DATASET ID	PURPOSE
raw_google_ads	Google Ads managed transfer
raw_meta_ads	Meta Ads managed transfer
raw_shopify	Shopify managed transfer
analytics_staging	Cleaned source-level views
analytics_marts	Dashboard-ready reporting views

PHASE 2 - GA4

Record the correct GA4 property and web stream

Do not link the wrong property. Confirm the website stream contains raphandremy.com.

YOU

OPEN THIS PAGE

https://analytics.google.com/
Use the property selector at the top to select the Raph & Remy GA4 property.

1. **Click Admin**
   Use the gear icon in the lower-left area.
2. **Open Property settings**
   Copy the numeric Property ID into the worksheet.
3. **Open Data streams**
   Under Data collection and modification, select Data streams.
4. **Open the Web stream**
   The website URL must be www.raphandremy.com or the approved canonical domain.
5. **Copy the Measurement ID**
   Record the value beginning with G-.
6. **Check access**
   Under Property access management, your account must show Editor or Administrator.

DO NOT CONTINUE UNTIL

CHECK 1: The property name is correct.
CHECK 2: The web stream URL is correct.
CHECK 3: Property ID and Measurement ID are recorded.

PHASE 2 - GA4

Validate ecommerce tracking before exporting it

BigQuery cannot repair missing or duplicated website events. Test the live store first.

YOU

OPEN THIS PAGE

https://analytics.google.com/
Open Admin > Data display > DebugView in one tab.

1. **Open the store in another tab**
   Use a browser with Google Tag Assistant or your approved debug method enabled.
2. **View a product**
   DebugView should receive view_item once.
3. **Add the product to cart**
   DebugView should receive add_to_cart once.
4. **Begin checkout**
   DebugView should receive begin_checkout once.
5. **Complete a controlled test order**
   Use a test-payment method approved by the store owner.
6. **Open the purchase event**
   Confirm transaction_id, value, currency and items are present.
7. **Check duplicates**
   The same purchase must not appear twice.

STOP IF PURCHASE IS WRONG

Do not start modelling revenue until the purchase event contains the correct order ID, value, currency and product array. Give a DebugView screenshot to Codex and repair tracking first.

DO NOT CONTINUE UNTIL

CHECK 1: All four ecommerce events appear.
CHECK 2: Purchase value and currency match the test order.
CHECK 3: The purchase event appears once.

PHASE 2 - GA4

Link GA4 to BigQuery

Use Daily export for the MVP. Streaming is optional and creates additional cost and complexity.

YOU

OPEN THIS PAGE

https://analytics.google.com/
Select the correct property, then open Admin.

1. **Open Product links > BigQuery links**
   Click Link.
2. **Choose a BigQuery project**
   Select the Project ID recorded in the worksheet, then Confirm.
3. **Choose the location**
   Select the same approved location used by the other datasets.
4. **Choose the web stream**
   Include the Raph & Remy website stream. Do not exclude ecommerce events.
5. **Choose export frequency**
   Turn on Daily. Leave Streaming off for the first launch.
6. **Review**
   The project, location and web stream must all be correct.
7. **Click Submit**
   GA4 creates a dataset named analytics_PROPERTY_ID.

Official guide: Set up GA4 BigQuery export
https://support.google.com/analytics/answer/9823238

DO NOT CONTINUE UNTIL

CHECK 1: The BigQuery link shows Active in GA4.
CHECK 2: Daily export is enabled.

WAIT AND CHECK

Verify the first GA4 table

Google states that data should begin flowing within 24 hours. Do not repeatedly delete and recreate the link while waiting.

WAIT

OPEN THIS PAGE

https://console.cloud.google.com/bigquery
Expand the dashboard project in the Explorer panel.

1. **Look for analytics_PROPERTY_ID**
   The numeric portion must match the GA4 Property ID.
2. **Expand the dataset**
   Look for a table named events_YYYYMMDD.
3. **Open the table**
   Click Preview. Event rows should appear.
4. **Check yesterday**
   Daily export normally contains the previous day after processing.
5. **Record the first available date**
   GA4 native export does not backfill earlier event-level history.

IF NOTHING APPEARS AFTER 24 HOURS

Confirm billing, the BigQuery API, the GA4 link status and the firebase-measurement service account. Do not create a second GA4 property.

DO NOT CONTINUE UNTIL

CHECK 1: The analytics dataset exists.
CHECK 2: At least one events_YYYYMMDD table contains rows.

PHASE 2 - GOOGLE ADS

Find the correct Google Ads Customer ID

The managed BigQuery connector does not require a Google Ads developer token for standard reporting.

YOU

OPEN THIS PAGE

https://ads.google.com/
Open the Raph & Remy advertising account.

1. **Confirm the account name**
   Do not use a test account or unrelated client account.
2. **Copy the Customer ID**
   It is the ten-digit number shown near the account name, normally formatted 123-456-7890.
3. **Check whether an MCC is used**
   If one manager account should cover several ad accounts, record the manager Customer ID instead.
4. **Open Admin > Access and security**
   Confirm your Google account has at least read access.
5. **Confirm 2-Step Verification**
   Individual user authorization requires it.

NO DEVELOPER TOKEN

Choose the managed BigQuery transfer and Standard reports. Apply for an API developer token only if a required field cannot be delivered by the connector later.

DO NOT CONTINUE UNTIL

CHECK 1: The correct Customer ID is recorded.
CHECK 2: Your account has access and 2-Step Verification.

PHASE 2 - GOOGLE ADS

Create the Google Ads transfer

Use Standard reports, include Performance Max if it is used, and let Google refresh recent attribution changes.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/bigquery/transfers
Confirm the dashboard project before clicking anything.

1. **Click Create transfer**
   For Source, select Google Ads.
2. **Enter Customer ID**
   Use the recorded individual or manager Customer ID.
3. **Choose Standard report type**
   Do not create a custom Google Ads query for the MVP.
4. **Include Performance Max tables**
   Turn this on if Performance Max or Shopping campaigns are active.
5. **Set Refresh window**
   Choose 30 days so recent attribution changes are updated.
6. **Choose Dataset**
   Select raw_google_ads.
7. **Name the transfer**
   Use Raph Remy Google Ads Daily.
8. **Schedule and notifications**
   Choose daily, enable email failure notifications, then click Save and authorize.

Official guide: Google Ads BigQuery transfer
https://docs.cloud.google.com/bigquery/docs/google-ads-transfer

WAIT AND CHECK

Verify Google Ads and load historical dates

The first successful run proves the credentials and destination are correct. Add history only after that run succeeds.

CHECK

1. **Open BigQuery > Data transfers**
   Select Raph Remy Google Ads Daily.
2. **Open Run history**
   The latest run must show Succeeded.
3. **Inspect raw_google_ads**
   Campaign, AdGroup, Ad and performance tables or views should exist.
4. **Compare yesterday's spend**
   Compare the same date and account with the Google Ads interface.
5. **Schedule a backfill**
   From the transfer, choose Schedule backfill and start with the previous 90 days.
6. **Extend carefully**
   After 90 days succeeds, add older ranges needed for year-over-year reporting.

USE THE SAME DATE AND TIME ZONE

Minor differences can result from attribution updates. A large difference normally means the wrong Customer ID, date, currency or table grain.

DO NOT CONTINUE UNTIL

CHECK 1: The run status is Succeeded.
CHECK 2: Spend is within an explained tolerance.
CHECK 3: The required historical range is loaded.

PHASE 2 - SEARCH CONSOLE

Confirm the Search Console property and ownership

Bulk export can only be started by an Owner. Use the property that contains the complete domain.

YOU

OPEN THIS PAGE

https://search.google.com/search-console
Select the Raph & Remy property from the property selector.

1. **Choose the property**
   Prefer the Domain property if it covers raphandremy.com and all protocols and subdomains.
2. **Open Settings**
   Click Users and permissions.
3. **Confirm your permission**
   Your row must show Owner, preferably Verified owner.
4. **Record the property**
   Write the exact domain or URL-prefix value in the worksheet.
5. **Check recent performance**
   Open Performance > Search results and confirm clicks and impressions are present.

DO NOT CREATE A SECOND PROPERTY

If you are not an Owner, ask the verified owner to grant access. Creating a new property does not transfer the existing data or verification automatically.

PHASE 2 - SEARCH CONSOLE

Grant the export service account access to Google Cloud

Search Console writes data using a Google-managed service account. Enter the address exactly.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/iam-admin/iam

Confirm the dashboard project is selected.

1. **Click Grant access**
   A panel named Add principals opens.
2. **Enter the new principal**
   Paste search-console-data-export@system.gserviceaccount.com
3. **Add the first role**
   Select BigQuery Job User.
4. **Add another role**
   Select BigQuery Data Editor.
5. **Click Save**
   The special service account should now appear in the IAM list.

```
search-console-data-export@system.gserviceaccount.com
```

Official guide: Search Console bulk export setup
https://support.google.com/webmasters/answer/12917675?hl=en-GB

DO NOT CONTINUE UNTIL

CHECK 1: The principal appears in IAM.

CHECK 2: Both BigQuery roles are listed.

PHASE 2 - SEARCH CONSOLE

Start the daily Search Console export

Search Console creates its own dataset. The dataset name must begin with searchconsole.

YOU

OPEN THIS PAGE

https://search.google.com/search-console
Select the confirmed Raph & Remy property.

1. **Open Settings > Bulk data export**
   If this option is missing, recheck Owner permission.
2. **Enter Cloud project ID**
   Paste the Project ID, not the Project number.
3. **Enter the dataset name**
   Use searchconsole_raph_remy.
4. **Choose the location**
   Select the same approved BigQuery location.
5. **Click Continue**
   Review the project, dataset and location.
6. **Start the export**
   Confirm the setup and leave it active.

WHAT HAPPENS NEXT

Google states that the first export can take up to 48 hours. It starts from setup and does not automatically include earlier Search Console history.

DO NOT CONTINUE UNTIL

CHECK 1: Bulk data export shows Active.
CHECK 2: The Project ID and location match the worksheet.

WAIT AND CHECK

Verify the first Search Console tables

Wait for the first scheduled export. Do not change the dataset schema after Google creates it.

WAIT

OPEN THIS PAGE

https://console.cloud.google.com/bigquery
Expand searchconsole_raph_remy in the Explorer panel.

1. **Look for the dataset**
   It may take up to 48 hours to appear after successful configuration.
2. **Open ExportLog**
   The latest successful export should be recorded.
3. **Open searchdata_site_impression**
   Preview rows and confirm the property matches the store.
4. **Open searchdata_url_impression**
   Preview landing-page rows.
5. **Compare one completed date**
   Clicks and impressions should align with the Search Console Performance report.

DO NOT EDIT THE RAW SCHEMA

Adding or changing columns can break future exports. Build reporting views in analytics_staging instead.

DO NOT CONTINUE UNTIL

CHECK 1: ExportLog contains a successful date.
CHECK 2: Both impression tables contain rows.
CHECK 3: Clicks reconcile for one date.

PHASE 3 - META

Collect the Meta identifiers and permissions

Google Cloud calls the source Facebook Ads. It covers Facebook and Instagram advertising data.

YOU

OPEN THIS PAGE
https://business.facebook.com/settings
Select the Raph & Remy business portfolio.

1. **Open Business info**
   Copy the Business ID into the worksheet.
2. **Open Accounts > Ad accounts**
   Select the active account and record the Ad Account ID.
3. **Check your access**
   Your user must have access to the business and ad account.
4. **Open Events Manager**
   Record the Pixel or Dataset ID used by the website.
5. **Open Meta for Developers**
   Use developers.facebook.com and confirm your account can create or manage apps.

Business ID	_____ Safe to record
Ad Account ID	_____ Usually act_ plus numbers in API contexts
Pixel or Dataset ID	_____ Safe to record

PHASE 3 - META

Create the Meta Business app

The managed transfer needs a Meta App ID and App Secret. The app must belong to the correct business portfolio.

YOU

OPEN THIS PAGE

https://developers.facebook.com/apps/
Sign in with the Meta user that has business and ad-account access.

1. **Click Create App**
   Choose Business as the app type when asked.
2. **Name the app**
   Use Raph Remy BigQuery Reporting.
3. **Connect the business portfolio**
   Choose the same business that owns the ad account.
4. **Create the app**
   Complete any security confirmation shown by Meta.
5. **Open App Settings > Basic**
   Copy the App ID into the worksheet.
6. **Reveal the App Secret**
   Store it directly in a password manager. Do not put it in the worksheet or chat.
7. **Add Facebook Login for Business**
   From the app dashboard, choose Set up for this product.

IF META SHOWS REQUIRED ACTIONS

Open the Required actions section and complete the listed business verification or security tasks before continuing.

PHASE 3 - META

Start the transfer and copy its redirect URI

The transfer screen supplies a unique redirect URI. Add it to the Meta app before clicking Authorize.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/bigquery/transfers
Confirm the dashboard project, then click Create transfer.

1. **Choose Source: Facebook Ads**
   Do not choose a generic API connector.
2. **Choose Dataset**
   Select raw_meta_ads.
3. **Enter Client ID**
   Paste the Meta App ID.
4. **Enter Client secret**
   Retrieve the App Secret from the password manager. Do not save it in notes.
5. **Find the Redirect URI**
   Copy the URI shown beside or below the Refresh token field.
6. **Return to Meta for Developers**
   Open Facebook Login for Business > Settings.
7. **Add the URI**
   Paste it into Valid OAuth Redirect URIs, then click Save.

DO NOT CLICK SAVE IN GOOGLE CLOUD YET

Complete Meta authorization and the remaining transfer fields on the next page first.

PHASE 3 - META

Authorize Meta and configure the daily data

Start with a controlled set of objects and action collections. Too many combinations can trigger Meta rate limits.

YOU

1. **Return to the Google transfer form**
   Click Authorize beside Refresh token.
2. **Choose the Meta app**
   Select Raph Remy BigQuery Reporting and approve the requested account access.
3. **Select objects**
   Include AdAccounts, Campaigns, AdSets, Ads, AdCreatives, AdImages, AdInsights and AdInsightsActions.
4. **Limit authorized accounts**
   Turn on Fetch Data for Authorized Ad Accounts Only.
5. **Choose action collections**
   Start with Actions, ActionValues, CostPerActionType, OutboundClicks, OutboundClicksCtr, WebsitePurchaseRoas, VideoPlayActions, VideoP25WatchedActions, VideoP50WatchedActions, VideoP75WatchedActions and VideoP100_watchedActions.
6. **Set insights grain**
   Use Ad level, time increment 1 day and refresh window 30 days.
7. **Avoid breakdowns initially**
   Leave generic and action breakdowns empty for the first successful run.
8. **Schedule**
   Name it Raph Remy Meta Ads Daily, run daily, enable email failures and click Save.

Official guide: Facebook Ads BigQuery transfer
https://docs.cloud.google.com/bigquery/docs/facebook-ads-transfer

CURRENT LIMITATION
Google documents a minimum 24-hour frequency, a maximum 30-day refresh window and a 60-day expiry for long-lived user tokens.

WAIT AND CHECK

Verify Meta and schedule token renewal

Do not add placement breakdowns until the first ad-level transfer reconciles.

CHECK

1. **Open Data transfers**
   Select Raph Remy Meta Ads Daily and open Run history.
2. **Confirm Succeeded**
   If authorization fails, check the redirect URI, app role and ad-account access.
3. **Inspect raw_meta_ads**
   Confirm AdInsights, AdInsightsActions, Ads, AdCreatives and Campaigns contain rows.
4. **Compare yesterday's spend**
   Use the same ad account, date, time zone and attribution view in Ads Manager.
5. **Compare purchases and purchase value**
   Document the Meta attribution setting used by the dashboard.
6. **Set a renewal reminder**
   Create a reminder for day 50 to open the transfer, click Edit and authorize again.

DO NOT SUM BREAKDOWNS INTO CORE TOTALS

Placement, device and action-breakdown rows can repeat ad-level totals. Keep later breakdown transfers separate from the main paid-media totals.

DO NOT CONTINUE UNTIL

CHECK 1: The latest run succeeded.
CHECK 2: Spend and purchase values reconcile within an explained tolerance.
CHECK 3: A day-50 renewal reminder exists.

PHASE 4 - SHOPIFY

The Raph & Remy storefront is confirmed as Shopify

The public site loads Shopify storefront assets and Shopify-hosted scripts. Use the Shopify path in this guide.

YOU

PUBLIC SIGNAL	WHAT IT CONFIRMS
/products, /collections and /policies routes	Standard Shopify storefront structure
cdn.shopify.com assets	Shopify hosts storefront resources
shopifycloud storefront scripts	The live storefront uses Shopify infrastructure

ADMIN LOGIN STILL REQUIRED

The public platform check does not reveal the private shop name ending in .myshopify.com. Copy that value from Shopify Admin before creating the transfer.

1. **Open Shopify Admin**
   Sign in using the store-owner or approved staff account.
2. **Open Settings > Domains**
   Find the domain ending in .myshopify.com.
3. **Record only the shop name**
   If the domain is storename.myshopify.com, record storename.
4. **Record store currency and time zone**
   Open Settings > General. These definitions will be used during reconciliation.

PHASE 4 - SHOPIFY

Create the reporting app in Shopify's Dev Dashboard

Shopify stopped allowing new legacy custom apps from 1 January 2026. New apps are created through the Dev Dashboard.

YOU

OPEN THIS PAGE
Shopify Admin > Settings > Apps
Use the live Raph & Remy store.

1. **Click Develop apps**
   If the button is missing, your user needs the App development > Develop permission.
2. **Click Build apps in Dev Dashboard**
   The Shopify Dev Dashboard opens.
3. **Click Create app**
   Choose Start from Dev Dashboard.
4. **Name the app**
   Use Raph Remy BigQuery Reporting.
5. **Set the app URL**
   If no interface is required, Shopify permits the default https://shopify.dev/apps/default-app-home.
6. **Choose an API version**
   Leave the current stable Admin API version selected by Shopify.
7. **Continue to Access**
   The next page lists the API scopes.

Official guide: Create and install custom apps
https://help.shopify.com/en/manual/apps/installing-apps

NO CUSTOMER EXPORT FOR THE MVP
The dashboard needs aggregate sales, product and refund data. Do not request customer names, addresses, emails or phone numbers unless a separate approved use case requires them.

PHASE 4 - SHOPIFY

Configure the minimum reporting access

Start with read-only access. The app must not edit orders, customers, products or inventory.

YOU

READ SCOPE	WHY IT IS NEEDED
Orders and transactions	Revenue, discounts, order counts and payment values
Refunds and returns	Net revenue and return-rate reporting
Products and variants	Product names, types, SKUs and merchandising analysis
Inventory and locations	Optional stock and availability context
Fulfilments	Optional order-status and delivery analysis
Discounts	Promotion and discount reporting

```
read_orders, read_products, read_inventory, read_locations, read_fulfillments, read_returns, read_discounts
```

1. **Select the corresponding read-only Admin API scopes**
   Do not select any scope beginning with write_.
2. **Release the app version**
   Enter a version name such as Reporting v1, then click Release.
3. **Install the app**
   From the app's Installs section, choose the Raph & Remy store and click Install.
4. **Review permissions**
   The installation screen must show read-only data access.
5. **Record Client ID**
   Safe to store in the worksheet.
6. **Store Client Secret**
   Place it in the password manager. Do not paste it into chat.

OLDER ORDERS MAY REQUIRE APPROVAL

Access to orders older than Shopify's default API window can require additional read_all_orders approval. Launch the current-data MVP first, then request older history if required.

PHASE 4 - SHOPIFY

Create the Shopify BigQuery transfer

The managed Shopify connector is still a Preview feature. Use it for the MVP, monitor it closely and keep the raw data separate.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/bigquery/transfers
Confirm the dashboard project, then click Create transfer.

1. **Choose Source: Shopify**
   If Shopify is not listed, confirm the Data Transfer API and project location.
2. **Enter Shop name**
   Use only the portion before .myshopify.com.
3. **Enter Client ID and Client Secret**
   Retrieve the secret from the password manager.
4. **Browse objects**
   Select Orders, OrderTransactions, Refunds, Returns, Products, ProductVariants, Collections, InventoryItems, Locations, Fulfillments and Discounts objects available to the app.
5. **Avoid Customers**
   Do not select Customers for the dashboard MVP.
6. **Choose Dataset**
   Select raw_shopify.
7. **Name and schedule**
   Use Raph Remy Shopify Daily, choose daily and enable failure emails.
8. **Save**
   The connector automatically starts according to the schedule.

Official guide: Shopify BigQuery transfer
https://docs.cloud.google.com/bigquery/docs/shopify-transfer

PREVIEW STATUS

Google currently states that the Shopify connector is Preview and has no transfer charge while in Preview. BigQuery storage and query charges can still apply.

WAIT AND CHECK

Verify Shopify revenue, refunds and products

Shopify is the source of truth for business revenue. GA4 and ad-platform revenue are attribution views, not additive sales totals.

CHECK

1. **Open Data transfers**
   Select Raph Remy Shopify Daily and confirm the latest run is Succeeded.
2. **Inspect raw_shopify**
   Orders, OrderTransactions, Refunds, Products and ProductVariants must contain rows.
3. **Choose one completed seven-day period**
   Avoid today and partially processed orders.
4. **Compare order count**
   Match the Shopify Analytics or Orders report for the same status and date definition.
5. **Compare gross sales, discounts and refunds**
   Document any excluded test, cancelled or unpaid orders.
6. **Calculate net revenue**
   Use one agreed formula and record it in the reporting definitions.

EXECUTIVE REVENUE DEFINITION

Use Shopify net revenue for executive reporting. Show GA4-, Google- and Meta-attributed revenue separately and never add them together.

DO NOT CONTINUE UNTIL

CHECK 1: The transfer succeeded.
CHECK 2: Orders and net revenue reconcile for seven completed days.
CHECK 3: Refund and test-order treatment is documented.

PHASE 5 - CODEX

Prepare the prototype for live data without changing its design

Codex should first document the existing interface and create a reversible working structure. It must not invent API fields.

CODEX

PASTE INTO CODEX

Inspect the approved Raph & Remy dashboard prototype and its untouched backup. Preserve the existing Inter typography, branding, navigation and page hierarchy. Do not redesign anything. Create a DATA_CONTRACT.md file that lists every visible metric, chart, table, filter and opportunity card. For each item, record its current dummy-data location, required date grain, filter inputs and proposed API endpoint. Do not write API code yet. Show me the data contract and any ambiguous metric definitions.

CODEX MUST PRODUCE	YOU CHECK
Dashboard page inventory	Every visible page and component is listed
Dummy-data map	No demonstration value is missed
Endpoint plan	Each page requests only the data it needs
Definition questions	Revenue, ROAS, CPA and attribution are unambiguous

DO NOT CONTINUE UNTIL

CHECK 1: DATA_CONTRACT.md exists.
CHECK 2: The visual prototype is unchanged.
CHECK 3: Every dummy-data block has a future live source.

PHASE 5 - BIGQUERY

Collect the real table and column names for Codex

Connector schemas change. Codex must inspect what actually arrived rather than guessing column names.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/bigquery
Click the plus sign to create a new SQL query.

Replace YOUR_PROJECT_ID with the Project ID from the worksheet. Run the query once for each raw dataset by changing raw_google_ads to raw_meta_ads, raw_shopify and searchconsole_raph_remy.

```
SELECT table_name, column_name, data_type
FROM `YOUR_PROJECT_ID.raw_google_ads.INFORMATION_SCHEMA.COLUMNS`
ORDER BY table_name, ordinal_position;
```

1. **Click Run**
   BigQuery displays one row per column.
2. **Save the results**
   Use Save results > CSV for each dataset.
3. **Do not share secret values**
   Schema files contain names and types, not credentials.
4. **Attach the schema CSV files to Codex**
   Also tell Codex the GA4 dataset name analytics_PROPERTY_ID.

DO NOT CONTINUE UNTIL

CHECK 1: Schema files exist for every source.
CHECK 2: No dataset name or column name is being guessed.

PHASE 5 - CODEX

Ask Codex to create the BigQuery reporting model

Codex creates SQL files, tests and run instructions. It does not execute or deploy anything until you review the plan.

CODEX

PASTE AFTER ATTACHING THE SCHEMA FILES

Use the attached BigQuery schema exports and DATA_CONTRACT.md. Do not invent columns. Create idempotent BigQuery SQL files under a new sql folder. Build analytics_staging views for GA4 ecommerce, Google Ads daily performance, Meta ad and creative performance, Shopify orders and refunds, and Search Console query and landing-page performance. Then build analytics_marts views for executive summary, paid performance, creative intelligence, product funnel, organic search, opportunities and data health. Use Shopify net revenue as executive revenue. Keep Google, Meta and GA4 attributed revenue separate. Add date and source filters, safe division, duplicate-grain tests and a README with the exact run order. Do not execute the SQL. Show me every assumption first.

REVIEW THE ASSUMPTIONS

If Codex asks which purchase action, order status, currency or attribution window to use, answer from the source reconciliations. Do not tell it to choose automatically.

DO NOT CONTINUE UNTIL

CHECK 1: The sql folder and run-order README exist.
CHECK 2: No secret or customer PII is selected.
CHECK 3: Source revenue definitions remain separate.

REPORTING DEFINITIONS

Approve the metrics before running the SQL

The dashboard should state which system owns each number. This prevents conflicting revenue and ROAS claims.

YOU

DASHBOARD METRIC	SOURCE OF TRUTH	USE
Net revenue	Shopify orders minus agreed refunds and discounts	Executive performance
Orders	Shopify completed or paid orders	Business outcome
Sessions and funnel	GA4 ecommerce events	Website behaviour
Google spend and ROAS	Google Ads transfer	Google campaign optimisation
Meta spend and ROAS	Meta transfer	Meta campaign optimisation
Paid blended efficiency	Paid spend divided by Shopify revenue	Budget allocation context
Organic clicks and CTR	Search Console bulk export	SEO opportunity discovery

Reporting time zone	_____ Recommended: store time zone
Reporting currency	_____ Expected: SGD
Included order statuses	_____ Write the exact rule
Refund date treatment	_____ Order-date restatement or refund-date adjustment
Default comparison	_____ Previous period and previous year

PHASE 5 - CHECK

Run the SQL in order and reconcile every mart

Use Codex's run-order README. Run one file at a time in BigQuery Studio and stop at the first error.

CHECK

1. **Open the first SQL file**
   Copy it from the workspace into a BigQuery query tab.
2. **Replace approved placeholders**
   Project ID and dataset names must match the worksheet.
3. **Review bytes processed**
   BigQuery shows an estimate before execution. Stop if it is unexpectedly large.
4. **Click Run**
   Copy any error message back to Codex without credentials.
5. **Continue in README order**
   Create staging views before mart views.
6. **Run the included tests**
   Duplicate-grain and nonnegative-metric tests must pass.
7. **Compare each mart**
   Use one completed seven-day period and the corresponding source interface.

IF A TEST FAILS

Do not change the raw tables. Paste the SQL filename, complete error and relevant schema rows into Codex. Ask it for the smallest correction and rerun only that file.

DO NOT CONTINUE UNTIL

CHECK 1: All SQL files ran successfully.
CHECK 2: All tests pass.
CHECK 3: Shopify revenue and platform spend reconcile.

PHASE 6 - CODEX

Build a private server that reads BigQuery

The browser must never receive a Google key. Cloud Run will use its own service identity to query approved mart views.

CODEX

PASTE INTO CODEX

Using DATA_CONTRACT.md and the analytics_marts view names, convert the dashboard project into a production-ready Node.js service for Google Cloud Run. Use the official @google-cloud/bigquery client and Application Default Credentials. Do not create, download or reference a service-account JSON key. Add read-only API routes for executive summary, paid performance, creative performance, products, organic search, opportunities and data health. Validate date ranges and source filters, cap row counts, return a last_refreshed timestamp and use parameterized queries. Preserve a local mock-data mode so the interface can still be previewed without cloud credentials. Add automated tests and a README. Do not deploy yet.

REQUIRED ROUTE	PRIMARY VIEW
/api/executive-summary	mart_executive_summary
/api/paid-performance	mart_paid_performance
/api/creative-performance	mart_creative_performance
/api/products	mart_product_funnel
/api/organic-search	mart_organic_search
/api/opportunities	mart_opportunities
/api/data-health	mart_data_health

PHASE 6 - CODEX

Replace the dummy data page by page

Connect one page, preview it and test it before moving to the next page.

CODEX

PASTE INTO CODEX

Connect only the Executive Summary page to /api/executive-summary. Preserve the approved layout and Inter typography. Add a visible loading state, a professional empty state, an error message with a Retry button and a data-freshness label. Keep the other pages on dummy data for now. Run the existing tests and start the local preview. Tell me exactly which values are live and how I can compare them with BigQuery.

1. **Preview Executive Summary**
   Check desktop and narrow-screen layouts.
2. **Compare seven completed days**
   Every card must match the executive mart.
3. **Test date filters**
   The page and API must use the same inclusive dates.
4. **Test failure state**
   Use Codex's mock failure switch or temporarily disable the API route.
5. **Repeat page by page**
   Paid, Creative, Products, Organic, Opportunities and Data Health follow only after the prior page passes.

DO NOT CONNECT EVERYTHING AT ONCE

A single incorrect page is easy to diagnose. Seven incorrect pages make it difficult to know whether the problem is the source, SQL, API or interface.

LOCAL QUALITY CHECK

Test the complete dashboard before cloud deployment

The dashboard is ready to deploy only when the local preview works in live-data mode and in failure conditions.

CHECK

CHECK	PASS CONDITION
Navigation	Every menu item opens the intended page and the hierarchy is consistent
Date controls	7-, 28-, 30- and 90-day ranges use completed dates consistently
Loading	No blank cards or layout jumps while data loads
Empty data	A clear no-data state replaces misleading zeros
Errors	Retry works and no secret, SQL or stack trace appears
Freshness	Every page shows the latest source refresh
Responsive layout	Cards, tables and navigation remain usable on a narrow screen
Performance	Pages use mart views and do not return unnecessary columns
Accessibility	Keyboard focus, contrast and labels are usable

ASK CODEX TO VERIFY

Run the full test suite and production build. Report every failure with its filename and likely cause. Do not change the design to make a test pass. After the build succeeds, inspect the rendered HTML for dummy values, exposed secrets and broken routes.

PHASE 7 - CLOUD RUN

Create the dashboard service account

Cloud Run uses this identity to query BigQuery without a downloaded key file.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/iam-admin/serviceaccounts
Confirm the dashboard project.

1. **Click Create service account**
   Name it dashboard-runtime.
2. **Create and continue**
   Do not create a key.
3. **Grant BigQuery Job User**
   This allows the service to run approved queries.
4. **Grant BigQuery Data Viewer**
   Use project-level access for the first private MVP, then reduce to analytics_marts if your policy requires.
5. **Finish**
   Record the email ending in @PROJECT_ID.iam.gserviceaccount.com.
6. **Inspect Keys**
   The Keys tab should show no user-managed keys.

NO JSON KEY

Google recommends a user-managed service identity for Cloud Run and warns not to set GOOGLE_APPLICATION_CREDENTIALS on the service. Attach the service account to Cloud Run instead.

Official guide: Cloud Run service identity
https://docs.cloud.google.com/run/docs/securing/service-identity

PHASE 7 - CODEX

Prepare the exact deployment command

Codex should verify the build and generate a command using the Project ID, chosen region, dashboard-runtime identity and Identity-Aware Proxy.

CODEX

PASTE INTO CODEX

Prepare this project for Google Cloud Run source deployment. Confirm the application listens on process.env.PORT and that the production build starts correctly. Use service name raph-remy-growth-dashboard, the Project ID and region I provide, and the dashboard-runtime service account. Generate one gcloud run deploy command that keeps unauthenticated access disabled and enables Identity-Aware Proxy using --iap. Do not run the command or change Google Cloud until I confirm. Also list the APIs and IAM roles the deployer may need.

WHAT YOU SHOULD RECEIVE

Codex should return a final command containing --source ., --region, --project, --service-account, --no-allow-unauthenticated and --iap.

VERIFY IN THE COMMAND	EXPECTED VALUE
Service	raph-remy-growth-dashboard
Project	Your recorded Google Cloud Project ID
Region	Your approved Cloud Run region
Identity	dashboard-runtime@PROJECT_ID.iam.gserviceaccount.com
Public access	Disabled; IAP enabled

PHASE 7 - DEPLOY

Run the approved deployment in Google Cloud Shell

Cloud Shell includes the Google Cloud command-line tools. You only paste the command that Codex prepared and you reviewed.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/

Confirm the dashboard project, then click the Cloud Shell terminal icon in the top bar.

1. **Open Cloud Shell**
   Wait until the prompt appears at the bottom or in a separate window.
2. **Upload the project archive**
   Use the Cloud Shell menu > Upload and select a ZIP prepared by Codex, then unzip it using the exact command Codex provides.
3. **Change into the project folder**
   Ask Codex for the exact cd command if the folder name differs.
4. **Paste the reviewed deploy command**
   Answer y only when Google asks to enable a named required API.
5. **Wait for success**
   Cloud Build may take several minutes. Do not close the shell.
6. **Copy the service URL**
   The final output shows a run.app URL.
7. **Test authentication**
   Open the service URL. IAP should request Google sign-in instead of exposing the dashboard.

Official guide: Deploy Cloud Run from source
https://docs.cloud.google.com/run/docs/deploying-source-code

IF THE BUILD FAILS

Copy the last complete error block to Codex. Do not repeatedly rerun the same command or enable unrelated roles.

PRIVATE ACCESS

Add only the people who should open the dashboard

Identity-Aware Proxy provides the browser sign-in. Start with yourself and the CEO. Add clients only after stable refreshes.

YOU

OPEN THIS PAGE

https://console.cloud.google.com/run
Open the raph-remy-growth-dashboard service.

STEP	ACCESS ACTION
1	Open the service Security tab Confirm Require authentication and Identity-Aware Proxy are selected.
2	Open the IAP policy Click Edit policy or Manage access for the IAP-secured service.
3	Enter an approved Google account email Start with your own account and the CEO's account.
4	Assign IAP-Secured Web App User Do not grant Editor, Owner or administrator roles to dashboard viewers.
5	Save and open the service URL IAP should show Google sign-in and then the dashboard for an approved account.
6	Test an unapproved session Use a separate account or signed-out browser. The dashboard must not load.
7	Record the access list Keep a simple owner-approved list and review it monthly.

CLIENT EXPERIENCE

Users outside the company's Google organization can require an external OAuth configuration. Keep the first review internal. Treat external client login as a separate access step rather than making the service public.

DO NOT CONTINUE UNTIL

CHECK 1: IAP is enabled.
CHECK 2: Approved users can sign in and unapproved users cannot access it.
CHECK 3: No source credentials are present in the browser.

OPPORTUNITY CENTRE

Add transparent opportunity rules after data is stable

Every recommendation must show the evidence, time range, minimum sample and suggested next action.

YOU

OPPORTUNITY	MINIMUM SIGNAL	SUGGESTED ACTION
Budget scale	Stable spend, enough conversions, ROAS above target and CPA below target	Increase budget 10 to 15 percent, then monitor
Creative winner	Meaningful spend with above-average CTR and purchase efficiency	Produce controlled variants
Creative fatigue	Frequency rises while CTR falls or CPA rises	Rotate the concept or audience
Landing-page test	Paid clicks are strong but add-to-cart rate is weak	Test offer, proof or page hierarchy
Product opportunity	High product views with weak add-to-cart rate	Test merchandising or product detail
Organic CTR	High impressions, positions 4 to 15 and weak CTR	Test title and description

ASK CODEX TO IMPLEMENT

Using the approved opportunity definitions, create rule-based SQL in analytics_marts.mart_opportunities. Every row must include opportunity_type, evidence_metric, current_value, comparison_value, date_range, minimum_sample_passed, confidence_label, suggested_action and status. Do not allow any rule to change budgets or publish ads.

A/B TESTING

Turn opportunities into controlled experiments

The dashboard should manage test decisions, not only display a list of ideas.

YOU

FIELD	WHAT YOU RECORD
Test ID	Stable ID such as CRO-001 or CREATIVE-001
Hypothesis	Why the change should affect behaviour
Control and variant	Exactly what each audience sees
Primary metric	One result used to decide the winner
Guardrail metrics	Metrics that must not materially worsen
Minimum sample	Required users, clicks, orders or conversions
Start and end	Planned dates and actual dates
Decision	Implement, iterate, reject or inconclusive

ASK CODEX TO ADD THE REGISTER

Create an A/B test register for the dashboard using the listed fields. Store only test definitions, aggregated results and decisions. Add backlog, planned, running, complete and stopped statuses. Do not calculate a winner when minimum sample has not passed, and never choose the most favourable metric after the test ends.

ONE PRIMARY METRIC

Choose the decision metric before the test launches. This prevents an inconclusive result from being presented as a win.

OPTIONAL AFTER LAUNCH

Add AI analysis only after the rules reconcile

AI should explain evidence and propose investigations. It must not control budgets, launch ads or change the dashboard's source metrics.

YOU

AI MAY	AI MUST NOT
Summarise a completed reporting period	Receive customer names, emails, addresses or order notes
Compare products, campaigns and creatives	Receive Meta or Shopify secrets
Explain which rule triggered an opportunity	Write directly to advertising accounts
Draft an experiment hypothesis	Declare a winner without the pre-agreed sample
Suggest questions for investigation	Replace the deterministic source-of-truth metrics

FUTURE CODEX PROMPT

Add a provider-neutral AI analysis service that receives only aggregated analytics_marts data. Store the provider key in Google Secret Manager and grant only the Cloud Run service identity access. Require every response to return the date range, supporting metrics, triggered rule and a draft status. Human approval is required before a recommendation appears as an action.

NOT PART OF THE FIRST GO-LIVE

Launch the live dashboard and complete several reconciled refreshes before enabling this section.

OPERATIONS

Follow a simple maintenance routine

A live dashboard is reliable only when transfers, definitions, access and costs are checked regularly.

YOU

FREQUENCY	CHECKS
Every working day	Data Health page, failed-transfer email, last refresh and yesterday completeness
Every week	Reconcile paid spend, Shopify net revenue and order counts for completed dates
Every month	Review access list, Google Cloud billing, query cost and opportunity-rule quality
Every 50 days	Renew the Meta long-lived user token before the 60-day expiry
After metric changes	Update reporting definitions, SQL tests and dashboard labels together
After deployment	Run the full build, route and reconciliation checks

WEEKLY CHECK PROMPT

Review the dashboard project's data-health outputs for the last seven completed days. Report failed or late sources, unusual row-count changes, reconciliation differences and approaching Meta token renewal. Do not change code, transfers or definitions. Give me a short ordered checklist of what to inspect.

TROUBLESHOOTING

Use the first failing checkpoint to find the problem

Do not rebuild an entire source connection when only a reporting view or dashboard page is wrong.

YOU

WHAT IS WRONG	FIRST PLACE TO CHECK	GIVE CODEX
Source dataset missing	Source link, permissions, billing and location	Screenshot of setup status
Transfer failed	BigQuery transfer Run history	Transfer name, time and full error
Raw totals wrong	Source account, date, currency and grain	Comparison table without secrets
Mart totals wrong	SQL tests and joins	SQL filename and source totals
API totals wrong	Route parameters and selected mart	Request URL and response without secrets
Page totals wrong	Date filters and frontend mapping	Screenshot and API response
Dashboard unavailable	Cloud Run revision and logs	Latest error block

SAFE SUPPORT REQUEST

Include the source name, affected date, expected value, actual value, transfer run time and exact error. Remove tokens, secrets, customer data and payment information.

GO-LIVE CHECKLIST

Launch only when every box is true

A working URL is not enough. Data accuracy, privacy, failure handling and access must all pass.

CHECK

AREA	FINAL CHECK
Prototype	Approved design is present and no unintended redesign was introduced
Google sources	GA4, Google Ads and Search Console refresh automatically
Meta	Transfer succeeds, values reconcile and renewal reminder exists
Shopify	Orders, refunds and net revenue reconcile
Definitions	Revenue, attribution, currency, time zone and refunds are documented
SQL	All model and duplicate-grain tests pass
Interface	No dummy values remain and all states are usable
Security	No keys in HTML, source control, chat or browser responses
Access	Only approved Google accounts can open the service
Cost	Budget alerts and billing review are active
Operations	Data Health and weekly reconciliation routine exist

LAUNCH DECISION

Start with yourself and the CEO. Add client users only after the dashboard has completed several successful daily refreshes without unexplained reconciliation differences.

FIRST 30 DAYS

Expand only after the live foundation is reliable

Use completed, reconciled data to decide what to add next.

YOU

PERIOD	FOCUS
First successful day	Confirm source freshness, dashboard access and no dummy values
First complete week	Reconcile spend, orders, refunds and revenue
Second week	Review opportunity rules and remove noisy recommendations
Third week	Launch one controlled creative or landing-page test
Fourth week	Review costs, user access and the Meta renewal date
After stability	Consider AI analysis, historical Search Console loading and advanced attribution

Your next action

ADD THE APPROVED PROTOTYPE FILE

The current project folder does not contain the dashboard prototype. Add that file first, then use the exact Codex prompt on page 4. Do not begin API coding before Codex confirms the prototype and backup.

REFERENCE

Plain-English glossary

These terms appear in the platforms and Codex prompts.

YOU

TERM	MEANING
API	A controlled way for one system to request data from another
BigQuery	Google Cloud's database for storing and analysing reporting data
Dataset	A folder containing related BigQuery tables and views
Transfer	An automatic scheduled copy from a source into BigQuery
Raw table	Connector data kept in its original structure
Staging view	A cleaned, standardised version of one source
Mart	A reporting view built for one dashboard decision
Backfill	Loading older dates outside the current refresh window
Service account	A non-human Google identity used by Cloud Run
OAuth	A secure approval flow used to authorize an account
Attribution	The rule that assigns a sale to a marketing interaction
Reconciliation	Comparing dashboard totals with the original platform

OFFICIAL SOURCES

Documentation used for this guide

Platform interfaces and connector limits can change. Open these sources if a label or option differs.

YOU

Official guide: GA4 BigQuery export
https://support.google.com/analytics/answer/9823238

Official guide: GA4 BigQuery schema
https://support.google.com/analytics/answer/7029846

Official guide: Google Ads BigQuery transfer
https://docs.cloud.google.com/bigquery/docs/google-ads-transfer

Official guide: Search Console bulk export
https://support.google.com/webmasters/answer/12917675?hl=en-GB

Official guide: Search Console export monitoring
https://support.google.com/webmasters/answer/12919198?hl=en

Official guide: Facebook Ads BigQuery transfer
https://docs.cloud.google.com/bigquery/docs/facebook-ads-transfer

Official guide: Shopify BigQuery transfer
https://docs.cloud.google.com/bigquery/docs/shopify-transfer

Official guide: Shopify custom apps
https://help.shopify.com/en/manual/apps/installing-apps

Official guide: Google Cloud budgets
https://docs.cloud.google.com/billing/docs/how-to/budgets

Official guide: Cloud Run deployment
https://docs.cloud.google.com/run/docs/deploying-source-code

Official guide: Cloud Run service identity
https://docs.cloud.google.com/run/docs/securing/service-identity

Official guide: Cloud Run Identity-Aware Proxy
https://docs.cloud.google.com/run/docs/securing/identity-aware-proxy-cloud-run

DOCUMENTATION DATE

Instructions were checked against official platform documentation available on 10 August 2026.